
EMILY WEI

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EXPERIENCE

The LCF Group

AI Engineer Intern

Lake Success, NY

May 2025 - Aug 2025

- Built an automated document extraction and validation pipeline in Python, eliminating manual data entry and integrating structured outputs into Salesforce and PostgreSQL workflows
- Developed and deployed a production-grade Python REST API with modular service architecture, pytest/BDD test coverage, and structured error handling
- Designed and maintained Azure DevOps CI/CD pipelines to support automated testing and staged deployments across QA, UAT, and production environments
- Evaluated and benchmarked multiple document extraction approaches, defining accuracy, latency, and resource constraints to deliver a lightweight CPU-based parser without ML infrastructure

America Reads & Counts

Teaching Assistant

New York, NY

Oct 2024 - Feb 2025

- Instructed 25+ high school students in Computer Science, Calculus, Algebra II, and Geometry via small-group and 1:1 tutoring
- Organized instructional materials and coursework for computer science topics, including problem sets and guided exercises
- Evaluated student assignments, providing detailed feedback to reinforce conceptual understanding and problem-solving skills

EDUCATION

New York University - Tandon School of Engineering

B.S. in Computer Science | Minors in Mathematics, Aerospace Engineering, Mechanical Engineering

New York, NY

May 2026

- **Relevant Coursework:** Algorithms, Applied Cryptography, Computer Networks, Databases, Machine Learning, Operating Systems, Software Engineering, Aerodynamics, Aerospace Propulsion, Compressible Flow, Flight Mechanics

PROJECTS

Wine Quality Classification ML Pipeline | [GitHub](#) | *Apr 2026 - May 2026*

New York, NY

- Engineered a modular machine learning pipeline in Python to classify wine quality from 6,500+ samples using Logistic Regression, SVM, KNN, PCA, and polynomial feature transformations
- Built reproducible preprocessing, training, and evaluation workflows with stratified train/validation/test splitting, train-only feature fitting, and automated comparison across 72 model configurations
- Developed reusable data-processing and model-training modules with scikit-learn, pandas, and NumPy, achieving a best test F1 score of 0.49 on an imbalanced classification task

Airline Reservation System | *Oct 2024 - Dec 2024*

New York, NY

- Designed and developed an airline reservation system with a Python/Flask backend, MySQL database, and REST-based architecture
- Deployed cloud-hosted infrastructure on AWS supporting core reservation workflows

Polynomial Arithmetic GUI (C++) | *Apr 2024 - May 2024*

New York, NY

- Implemented a polynomial arithmetic engine in C++ using linked lists to efficiently represent polynomials and support addition and multiplication operations
- Designed and integrated a graphical user interface to visualize polynomial operations and enable interactive user input beyond command-line execution
- Applied object-oriented design and dynamic memory management to ensure modularity and efficient resource handling

SKILLS & INTERESTS

Languages: Python, C/C++, SQL

Backend: REST APIs, Flask, FastAPI

Databases: PostgreSQL, MySQL

Cloud & DevOps: Docker, AWS (EC2, RDS, S3), Azure DevOps CI/CD

Tools: Git, GitHub, Linux, pytest

ML & NLP: Hugging Face, spaCy, LayoutLMv2, scikit-learn, pandas, NumPy

Interests: Space Exploration, Aerospace Technology, Hiking, Bouldering, Baking